Capstone Project: Machine Learning Approaches to Detection of Hate Speech

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1. Introduction

The development of social media platforms has significantly changed the face of digital communication, making it possible to exchange information and engage in public discourse like never before. However, this transformation has come with a major challenge: the proliferation of hate speech. Online hate speech, which describes language that targets individuals or groups based on their protected characteristics such as race, ethnicity, religion, gender, or sexual orientation, has become one of the biggest issues facing digital communication in the 21st century (UNESCO, 2023; Wilson & Land, 2021). Social media sites are now faced with the Herculean task of moderating millions of posts each day, and using algorithmic systems for this moderation is usually challenging, as accurately differentiating between actual hate speech and sarcastic, counter-speech, or offensive language is difficult (*Online Content Moderation*, 2025).

Lexical methods of detection using only the presence of specific terms are generally very imprecise because the system will identify any messages containing certain words as hate speech, regardless of context. This limitation, therefore, brings attention to the importance of developing better, more accurate methods to capture the nuances of the difference between hate and offensive content.

This study examines the performance of several machine learning models for identifying hate speech content from non-hateful content in the text of social media platforms. We use a comprehensive methodological framework that includes examination of traditional and deep learning approaches. For text representation, we will use the TF-IDF. We compare three TF-IDF configurations systematically: unigrams (single words), bigrams (word pairs), and a combination of unigrams and bigrams to see which of the feature extraction methods provides the best classification performance. For classification, we assess three-classical machine learning algorithms that have proven to be of high performance in text classification tasks: Logistic Regression, Support Vector Machine, and Multinomial Naive Bayes. Additionally, we compare these against DistilBERT, a transformer-based model that captures contextual word relationships (Devlin et al., 2019).

Furthermore, we also examine the classification from two points of view: a three-class classification, which separates hate speech, offensive language, and neutral content. The other approach is binary classification, which separates hate speech from all non-hate content (offensive and neutral).

1.1.Research Questions

The main research question studied in this report is: *How effectively can machine learning models (TF-IDF vs BERT) distinguish hate speech from non-hateful content in social media text, and which textual features are most predictive?*

To address this overarching question, we investigate the following subquestions:

a) Does BERT's contextual understanding outperform TF-IDF's frequency-based approach?
b) Which specific words or expressions act as 'anchors' for the TF-IDF model versus the BERT model when classifying a post as hateful?
c) Is there a correlation between low human annotator agreement (e.g., a 2-1 split) and machine misclassification? Does the model fail exactly where humans also struggle to agree?
d) How does the model accuracy change in binary (hate vs. non-hate) vs three class (hate vs. offensive vs. non-hate) categorization?

1.2.Significance and Relevance

The study of the detection of hate speech is at a central point in the field of communication science since it raises questions about the nature of harmful language in digital settings and how this can be effectively detected and mitigated. Understanding the elements of language that define hateful content has implications for theoretical frameworks on online discourse and has direct practical implications for content moderation strategies that social media platforms use. With millions of posts to be moderated each day, building accurate automation-based detection is not only a technical issue, but actually a must to protect users and keep online communities healthy.

Furthermore, this research adds to the already existing scholarly discussion in computational sciences about the capacities and limitations of text-based methods of detection (*Online Content Moderation*, 2025; Stoll et al., 2024). By contrasting traditional machine learning methods with deep learning models and analyzing the error in classification, we can gain insight into the types of hateful content for which existing automatic approaches are still proving computationally challenging, and in doing so, gain insight into the opportunities and limitations of existing automatic approaches. The comparison of methods based on TF-IDF and BERT also speaks to more general questions in computational social science about the trade-off

between model interpretability and performance, a consideration of particular importance when deploying systems for which freedom of expression is at stake.

2. Data Description

2.1 Data Source and Collection

In the present research, the Hate Speech and Offensive Language dataset is being used, which was initially gathered by Davidson et al. (2017). This dataset has established itself a leading source of information on the topic of automated hate speech detection. The dataset can be accessed publicly via Kaggle or in the original GitHub repository where the authors store the data.

A crowd-sourced hate speech lexicon offered on Hatebase.org was the method of data collection Davidson et al. (2017) used to gather tweets that include potentially hateful keywords. The resulting corpus is a perfect dataset to investigate the difficulty of the task of differentiating hate speech from other types of offensive or neutral language.

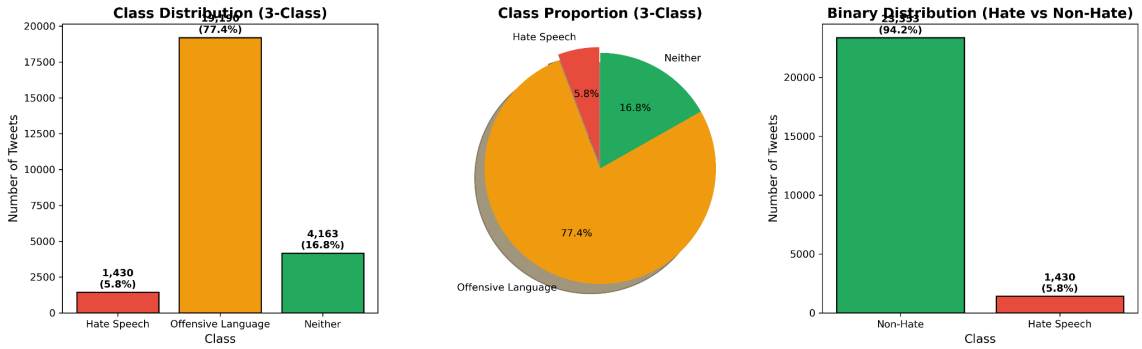
2.2 Dataset Characteristics and Class Distribution

The sample size is 24,783 tweets, with 3 to 9 crowd-sourced workers of CrowdFlower having annotated each of the tweets (M = 3.2 annotators). CrowdFlower, now Appen, is a crowdsourcing platform that recruits and manages freelancers to perform annotation tasks such as labeling text data (*About Appen*, n.d.). The length of an average tweet is 85.4 characters (14.1 words). The majority vote puts each tweet in one of the three categories: hate speech (class 0), offensive language (class 1), or neither (class 2).

The data has a strong imbalance of the classes. Hate speech represents 1430 tweets (5.77%), offensive language is the prevailing category with 19190 tweets (77.43%), and the neutral content constitutes 4163 tweets (16.80%). This distribution is comparable to the actual content of social media, in which explicit hate speech is not very common as opposed to overall offensive language (Vidgen et al., 2019). In binary classification, the offensive and neutral categories yield an even greater imbalance on the combination of the two categories, with 5.77% the hate speech and 94.23% the non-hate speech.

Figure 1

Class Distribution Graphs



3. Exploratory Data Analysis

3.1 Data Quality and Preprocessing

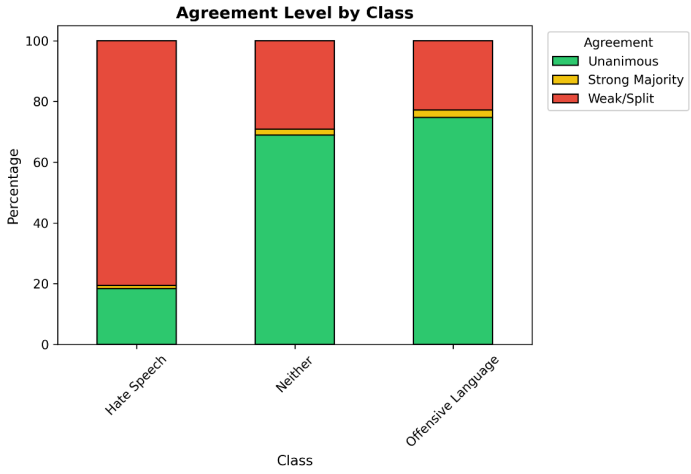
There are no missing values in the dataset, and there were three empty tweets after text preprocessing (lowercase conversion, URLs, user mentions, RT indicators, HTML entity, special characters, and whitespace normalization), and those were removed, resulting in 24,780 tweets. In the case of TF-IDF features, further stopword removal and lemmatization decreased the average word count (7.2 words) by 49% and maintained semantic content.

3.2.Annotator Agreement

The analysis of inter-annotator agreement also demonstrated a critical point, that hate speech has much lower agreement levels (only 18.4% unanimous) than offensive language (74.8% unanimous) or neither (69% unanimous). It indicates that even human beings attempt to detect hate speech consistently with ease, which puts an achievable limit on the model quality and puts into focus the ambiguous nature of differentiating between targeted hate speech and generic offensive material.

Figure 2

Annotator Agreement Bar Chart

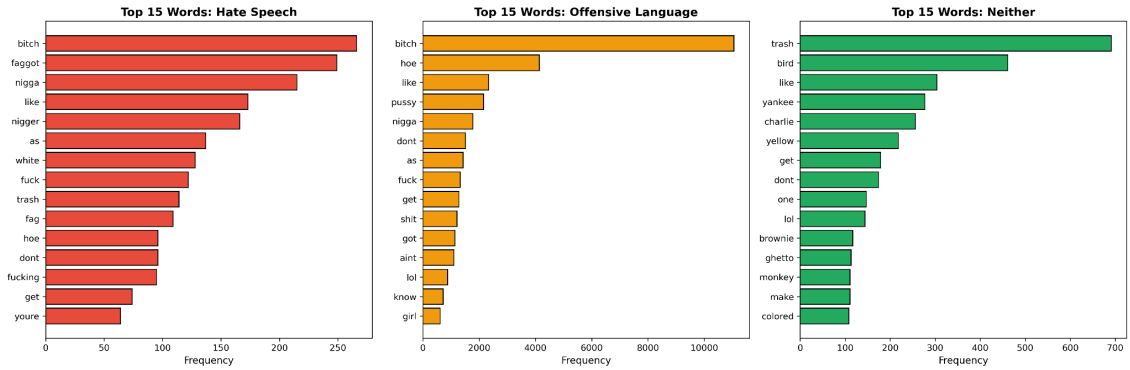


3.3.Vocabulary Analysis

The distinctive vocabulary patterns were analysed, and the obvious differences between classes were identified. Identity-targeting slurs (homophobic and racial epithets) are also inherent to hate speech, this category being found up to 7x to 40x more often than in other categories. The offensive language includes generic profanity, vulgar phrases not directed at any protected population, and social media slang. Interestingly, most of the 'neither' category is filled with dual-meaning words which could be used as slurs but are innocent in context (e.g., food terms such as brownie, sports terms such as Yankie). The fact that the presence of potentially offensive words does not always mean that the content is hate speech demonstrates a basic weakness of lexicon-based detection.

Figure 3

Top Words Bar Chart by Class



3.4.Key Findings

From the Exploratory Data Analysis, four useful pieces of information are gained and can be used for our modelling approach. The first is that the performance should be evaluated using F1-score due to severe imbalance between classes. Secondly, human consensuality in hate speech determination is poor and forms a realistic performance baseline. Third, clear vocabulary use justifies considering TF-IDF feature extraction. Fourth and last, the presence of contextual ambiguity because of the dual-meaning words encourages us to use context-aware models such as BERT.

4. Method

4.1.1 TF- IDF

TF-IDF (Term Frequency-Inverse Document Frequency) is a statistical technique that encodes text as numerical vectors. TF-IDF uses two variables (term frequency and inverse

document frequency) to give weight to words according to their frequency in a given document and their rarity in the overall corpus, respectively. The more often the words are used in the same document and the less in the dataset, the larger the weight.

We were driven to use TF-IDF as it is a well-established standard in text classification studies. In contrast to deep learning methods, TF-IDF can be interpreted with high accuracy, enabling the researcher to directly investigate what words or word combinations the model uses to make predictions. Transparency is especially important in the context of hate speech detection, in which the theoretical and practical implications of the knowledge of the particular linguistic features that distinguish hateful speech can be important.

There were three TF-IDF configurations that were used to answer the research question about the best feature representation. A unigram model that involves single words, a bigram model that involves two-word sequences, and a combined unigram and bigram model. The installation of this setup allows the comparison of the interaction of different settings with the classification task directly and the enhancement of phrasal context in detecting hate speech.

4.1.2 Model Training Process

The TF-IDF model preprocessing of text was performed in accordance with the traditional NLP standards. TF-IDF preprocessing included stopword removal, lemmatized words, and URLs, mentions and retweet markers.

Scikit-learn TfidfVectorizer was used to convert texts to numerical vectors. Three configurations were done: unigrams (10,000 features), bigrams (10,000 features), and combined unigrams-bigrams (15,000 features). Words that occur less than twice or more the 95% in documents were not included, and a sublinear scale of frequency of terms was used to reduce the effect of very frequent words.

The identical two labelling schemes were used: a three-class system (hate, offensive, neither) and a binary system (hate vs non-hate). The classifier used was Logistic Regression because it is effective when dealing with high-dimensional sparse data. To eliminate class imbalance, the class weights were adjusted to be balanced, whereby the misclassification of minority classes was automatically penalized more. The accuracy, F1-scores (macro and weighted), and confusion matrices were used to assess model performance.

4.2.1 DistilBERT

DistilBERT is a transformer-based model, specifically a “distilled” version of the bigger BERT model (Sanh et al., 2020). BERT is a state-of-the-art model for natural language processing (NLP), developed by researchers at Google in 2018. DistilBERT is a smaller and faster version of the introduced BERT-base model, with approximately 40% fewer parameters and a 60% faster pace, while still retaining 97% accuracy. This made it particularly suitable for our research settings with limited computational resources, while still providing state-of-the-art performance on nuanced language tasks such as hate speech detection.

Our motivation behind this choice stemmed from DistilBERT’s ability to detect the context. Transformer models such as DistilBERT differ from traditional text analysis methods (such as TF-IDF) in that they process language contextually. Rather than treating words as independent units, DistilBERT analyzes how words relate to one another bi-directionally within a sentence, allowing the model to capture meaning, tone, and implicit hostility. Similar to the TF-IDF process, two DistilBERT-based classification models were implemented for the fine-tuning process.

4.2.2 Model Training Process

Before model training, all tweets were preprocessed to remove noise introduced by the social media environment while preserving linguistic information relevant for classification. This cleaning process differed from the TF-IDF model significantly, as retaining each sentence’s natural linguistic form was significant for the DistilBERT model. Stopwords were retained to preserve sentence structure for contextual understanding. Thus, stopwords and grammatical structure were kept while the URLs, user mentions, and retweet markers were removed.

Later on, texts were converted into numerical representation using AutoTokenizer, which splits text into subword units. All inputs were padded to a fixed length of 64 tokens, motivated by exploratory data analysis (EDA) results, which indicated the majority of tweets in the dataset fell well below this length. This constraint reduced the computational cost while preserving nearly all content. Each input was represented by Input IDs, which map tokens to numerical values and attention masks, which indicate which tokens are meaningful or padding. These representations were packaged into a PyTorch dataset to ensure efficient data loading during training.

Three-class and binary classification setups were fine-tuned using a cross-entropy loss function, motivated by previous research methodology (Stoll et al., 2024). This loss penalizes

confident but incorrect predictions and is suited for imbalanced categorical outcomes such as hate speech detection. Fine-tuning was limited to 3 training epochs, a deliberate constraint intended to reduce overfitting and encourage the model to learn generalizable linguistic patterns rather than dataset-specific ones. Weight decay was applied to penalize overly complex parameter values, further improving generalization, particularly for the minority class, Hate. Mixed-precision training was enabled when supported by the hardware, improving training efficiency without compromising numerical stability. Model performance was evaluated using accuracy, F1-scores, and confusion matrices, with additional analysis of training and validation loss to assess generalization behavior.

5. Results

5.1 Subquestion a & d: Model Comparison and Classification Complexity

To evaluate the influence of feature extraction and contextual understanding on model performance, we analyzed confusion matrices and key evaluation metrics.

Figure 4

Confusion Matrices: TF-IDF (Unigram and Bigram)

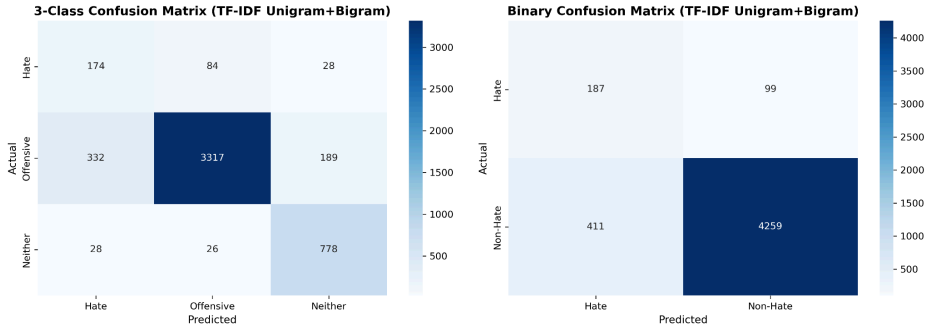


Figure 5

Confusion Matrices: DistilBERT

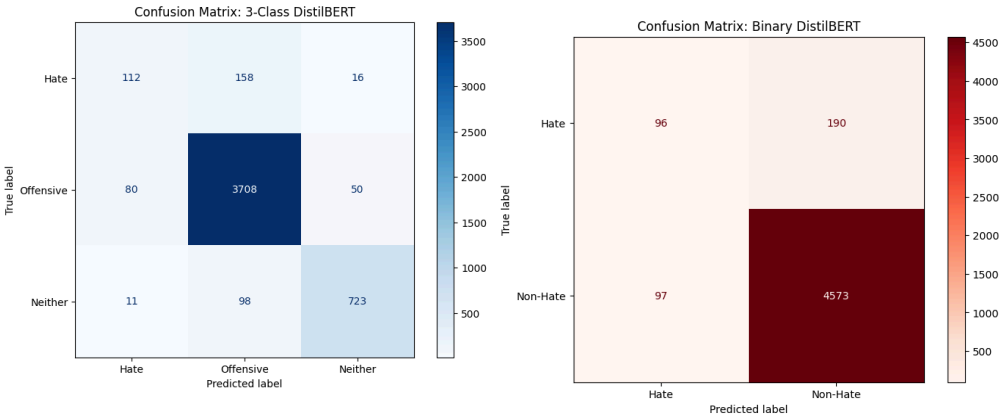


Table 1. Evaluation matrix for TF-IDF and DistilBERT models

	Task	Accuracy	F1 (Hate)	F1 (Offensive)	F1 (Neither)	F1 (Non-hate)
1	TF-IDF Binary (Unigram-Bigram)	.8995	.4315	—	—	.9449
2	DistilBERT Binary	.9421	.4008	—	—	.9696
3	TF-IDF 3-class (Unigram-Bigram)	.8606	.4249	.9120	.8512	—
4	DistilBERT 3-class	.9167	.4581	.9505	.8920	—

In the case of TF-IDF, three settings were used: unigrams, bigrams, and unigram-bigram. The combined structure was more effective than the other two methods, which gave lower accuracy and lower F1 on hate speech of three-classes (86.06% and 0.42 F1 Hate, respectively). This could signal that single-word pairs get lost in isolation, especially identity-targeting slurs, which are single-word lexical units. The binary TF-IDF model was more accurate (89.95) compared to its three-class equivalent because offensive and neutral categories are collapsed, making it simpler to determine the boundary between the classes.

For DistilBERT models, the binary model outperformed the three-class model by 2.5% in accuracy. However, the three-class model achieved a higher F1-score (0.46) for the Hate class, indicating that finer label resolution could improve minority-class sensitivity at the cost of overall stability. Notably, the F1-scores for the Hate class are low in both setups, with the models misclassifying nearly 60% of hate instances. This trade-off reflects a core challenge in hate speech detection: balancing overall robustness and precision against reliable recall for a minority and conceptually contested class. To explore the stronger model further, their generalization behvaior was also measured, which differed substantially. The three-class model showed a generalization gap of 0.13, indicating tendency of overfitting (Training Loss: 0.1684 vs. Validation Loss: 0.2984). The binary model exhibited a smaller gap of 0.063 (Training Loss: 0.1288 vs. Validation Loss: 0.1918), indicating a stronger generalization stability. This difference could indicate that increasing label granularity introduced learning instability, likely due to semantic overlap between the Hate and Offensive classes.

5.2 Subquestion b: Linguistic features drive model predictions

To identify which linguistic features act as strong anchors for model predictions, explainability analyses were conducted for both the TF-IDF and DistilBERT classifiers. While DistilBERT was examined using LIME explanations, TF-IDF predictions were analyzed through word frequency and distinctiveness patterns, translating its frequency-based decision mechanism.

For the TF-IDF model, predictive anchors are primarily high-frequency and class-distinctive lexical tokens. Word frequency analysis shows that the Hate Speech class is dominated by direct slurs and strongly derogatory terms, which occur disproportionately more often in this class than in the Offensive or Neither categories. The distinctiveness analysis further confirms that these terms appear more frequently in hateful posts relative to other classes, making them decisive anchors in TF-IDF's linear classification process. Because TF-IDF treats words as independent, the presence of such tokens alone can substantially increase the probability of a hateful prediction, regardless of the surrounding context. As a result, TF-IDF performs well for explicit, high-consensus hate speech, but remains open to misclassification when profane language is used casually or without hateful intent, such as offensive speech.

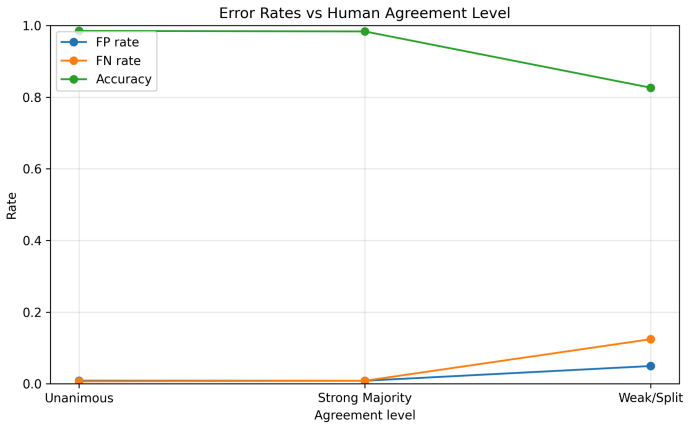
In contrast, DistilBERT relies on contextualized semantic presenters rather than singular word frequency. LIME-based explanations demonstrate that DistilBERT evaluates how words interact within a sentence to infer intent. In the hateful example *"*no ur wrong stupid n****r", the slur functions as the strongest anchor, but its influence is reinforced by surrounding hostile expressions such as "stupid" and "wrong". This indicates that BERT's prediction is driven by relational context rather than by the isolated presence of a single keyword. Conversely, in the non-hateful example "seeing so many basic b****s rn", LIME assigns negligible importance to the profane term *"*b****s*"*, and it confidently predicts a non-hateful label. This illustrates BERT's ability to suppress lexical toxicity when contextual cues signal colloquial or non-aggressive usage.

Overall, the explainability results signal a fundamental difference between the models: TF-IDF anchors predictions to statistically salient keywords, whereas BERT anchors predictions to contextual meaning and inferred intent. This distinction explains BERT's superior detection ability in ambiguous cases and its strong alignment with human judgments, particularly when casual profanity does not correspond to hateful intent.

5.3 Subquestion c: Human-machine agreement gap

Figure 6

Error Rates vs Human Agreement Level



In the figure above, you see the error rates of the model against the agreement level between the annotators. We can clearly see that whenever the annotators unanimously picked a label for a certain tweet, the model also picked correctly. However, when the annotators' opinions started deviating, leading to split agreement when labeling, a decrease in the model's accuracy was seen as well. This implies a linguistic ambiguity in the tweets, which suggests that both the model and humans work best with obvious hate.

The linguistic bias is caused annotators requiring extra context to be able to understand the true intention of the tweets. Some of the speech featured cultural references, jargon, or wordplay, while others contain certain phrases which could be understood in more than one way, depending on how the user meant it. When annotators feel unsure if something should be deemed hate or not, the model becomes unsure as well. All in all, it can be concluded that there is a linguistic bias present, and the model does not fail randomly. .

6. Conclusion

Our findings emphasize the various obstacles when attempting to detect hate speech. First off, the dataset itself should be balanced. In our research, the hate category had the least amount of entries, which limited how the model trained. Moreover, the tweets in the dataset used contain slang, jargon, wordplay, and cultural references that need further context in order to be understood correctly. The tweets analyzed are also quite short, which limits the model's and the annotators' ability to fully understand the context, which may lead to mislabelling.

Second, it was established that when the annotators did not agree, the model would produce more errors. If the annotators fail to agree on which category the tweet belongs to, the model struggles to create patterns, causing errors to occur.

Lastly, a big reason why there is confusion between the annotators on how to code a tweet is due to the fact that there is no clear description of the categories, which leads to confusion and mislabelling between the “hate” and “offensive” categories. Some tweets were found to use identity-related words that the model flagged as hate, however, the user was using the word in a sarcastic, non-hateful manner. The confusion between the categories illustrates why the binary classification outperforms three-class classification and thus creates more stability.

7. Discussion

Our research aimed to explore which of the models performed better at detecting hate in social media speech. We compared more traditional models, such as the TF-IDF model, against the DistilBERT model, in order to see which one will more effectively detect hate speech. Overall, DistilBERT performed better than the TF-IDF model, namely the binary DistilBERT model (hate vs non hate). This further supports the notion that linguistic ambiguity is a limitation, prior knowledge and context are important in interpreting hate, as the DistilBERT model is trained on the meaning of words instead of the frequency.

While the overall accuracy is high, the detection of hate speech is still tough. Low F1 scores were seen in the hate categories in all models, indicating that there are still issues with reliably detecting hate speech.

When compared to the three-class classification, the binary classification proved to be the more stable option, as the three-class classification introduced confusion between two of the categories, the offensive and the hate categories. The reason for this is the lack of conext and category descriptions for the tweets themselves. Providing a deeper understanding of how certain phrases or slurs were used in the hate speech tweets will provide annotators with a deeper understanding of what the user who wrote the tweet meant, as well as bring more clarity to which category the tweet truly belongs to.

7.1 Comparison of Davidson et al. (2017) study

Comparing our results with the foundational study by Davidson et al. (2017) reveals insights into the evolution of Natural Language Processing (NLP) methods in hate speech moderation. The 2017 study tested Logistic Regression, Naïve Bayes, Decision Trees, Random

Forests, and Linear SVMs, ultimately choosing Logistic Regression with L2 regularization. While the LR model achieved an overall accuracy of 91%, our binary DistilBERT model outperformed this, reaching an accuracy of 94.21%. This improvement suggests that modern LLMs provide a more reliable baseline for practical moderation. However, the detection of the "Hate" class remains a persistent challenge; Davidson et al. reported a "Hate" F1-score of 0.40, and our binary DistilBERT model achieved a similar score of 0.40.

7.2 Limitations and Future Research

7.2.1 Limitations

One of the main limitations of the study is the dataset used and the data collection method used in order to gather the tweets. The list of used tweets for this research was acquired with the use of a hate speech lexicon, meaning the tweets are labelled as hate for containing certain triggering profanity, even if the tweet itself is not intended to hate on anyone. At the same time, due to the triggering profanity in the dataset, the model has trouble picking up on subtle hate speech, which occurs when users replace offensive words with certain slang or another spelling variation of the word.

An additional limitation in the dataset is the imbalance between the categories. The amount of hate speech examples the model experiences during its training is too little, as the hate category has limited entries in the dataset, making it difficult for the model to create patterns that will help it distinguish hate. This explains why the F1 scores in the hate categories are low. Due to this, even when the model has a high accuracy score, it does not translate to absolute hate detection.

7.2.2 Future Research

In future studies, researchers should work with datasets that are balanced between categories and contain a higher number of entries. Using a larger and more balanced dataset will lead to better training in coding the tweets per category for both the model and the human annotators.

Moreover, additional context could be added to the model in order for it to get a deeper understanding of the situation surrounding the tweet. For example, looking into posts' threads (conversational replies), could support the model to capture tone and patterns of expression more effectively. This would lead to better detection and interpretation of sarcasm and irony, and would provide further insight into the situation where the tweet was commented on.

Lastly, the human annotators' training stands as a significant subject behind moderation research. The definition of categories, such as what groups as hate speech, should be adequately described for their classification task. Greater uniformity in annotator coding not only improves annotation reliability but also provides higher-quality training data, enabling machine learning models to better capture meaningful distinctions when learning and analyzing classification differences.

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